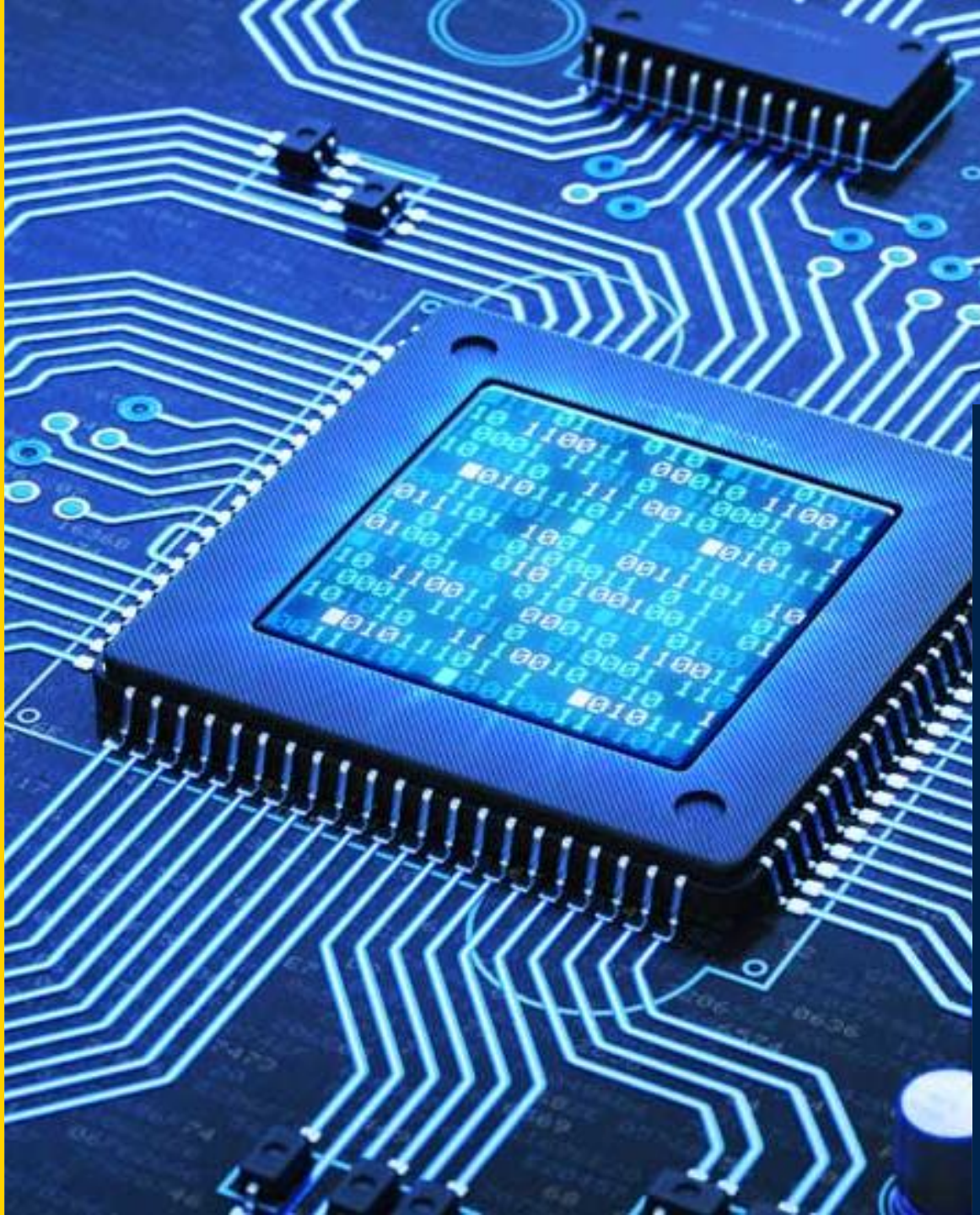


life.augmented

# Computer Architecture

## Practice Session 4

By: Eng. Abdulkareem Mohammed

Senior HW Digital Design Engineer

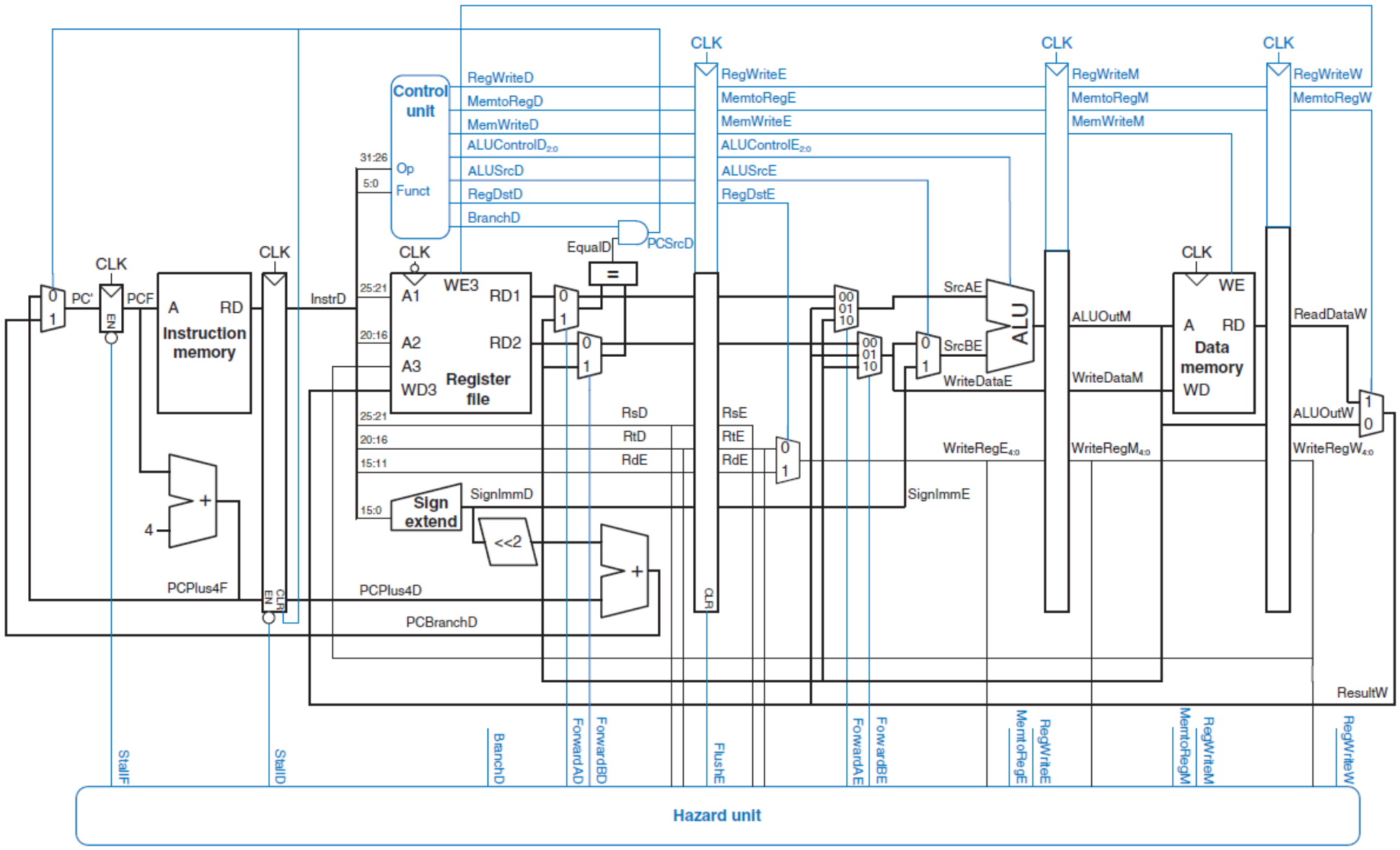
# Agenda

1. MIPS Basic Architecture
2. MIPS Hierarchy
3. Component Implementation
4. Testbench
5. Synthesis

# 1. MIPS Basic Architecture

life.augmented

# Introduction



# 2. MIPS Hierarchy

life.augmented

# 2. MIPS Hierarchy

- mips_tbench:
  - mips_top_inst
    - fetch_stage
      - instruction_mem_inst
      - pc_reg
      - f2d_regs
    - decode_stage
      - reg_file_inst
      - branch_calc_inst
      - sign_extend_inst
      - ctrl_unit_inst
      - d2e_regs
    - excute_stage
      - alu_inst
      - e2m_regs
    - mem_stage
      - mem_data_mem_inst
      - m2w_regs
    - hazard_unit

# 3. Component Implementation

life.augmented

# 2. Component Implementation

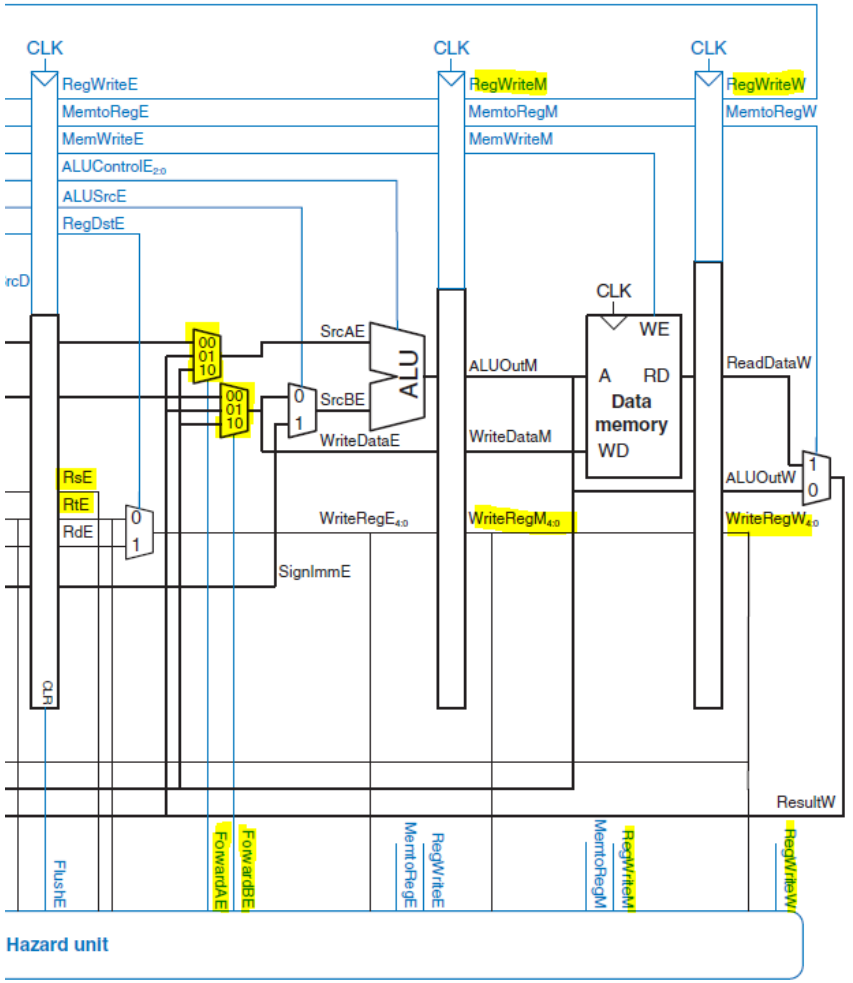
- hazard_unit Implementation:
  1. Data Hazard solved by forward in Execute

```
assign condition1_RsE = (RsE!=0) && (RsE == WriteRegM) && RegWriteM;
assign condition2_RsE = (RsE!=0) && (RsE == WriteRegW) && RegWriteW;

assign ForwardAE = (condition1_RsE)? 2'b10:(condition2_RsE)? 2'b01: 2'b00;


assign condition1_RtE = (RtE!=0) && (RtE == WriteRegM) && RegWriteM;
assign condition2_RtE = (RtE!=0) && (RtE == WriteRegW) && RegWriteW;

assign ForwardBE = (condition1_RtE)? 2'b10:(condition2_RtE)? 2'b01: 2'b00;
```

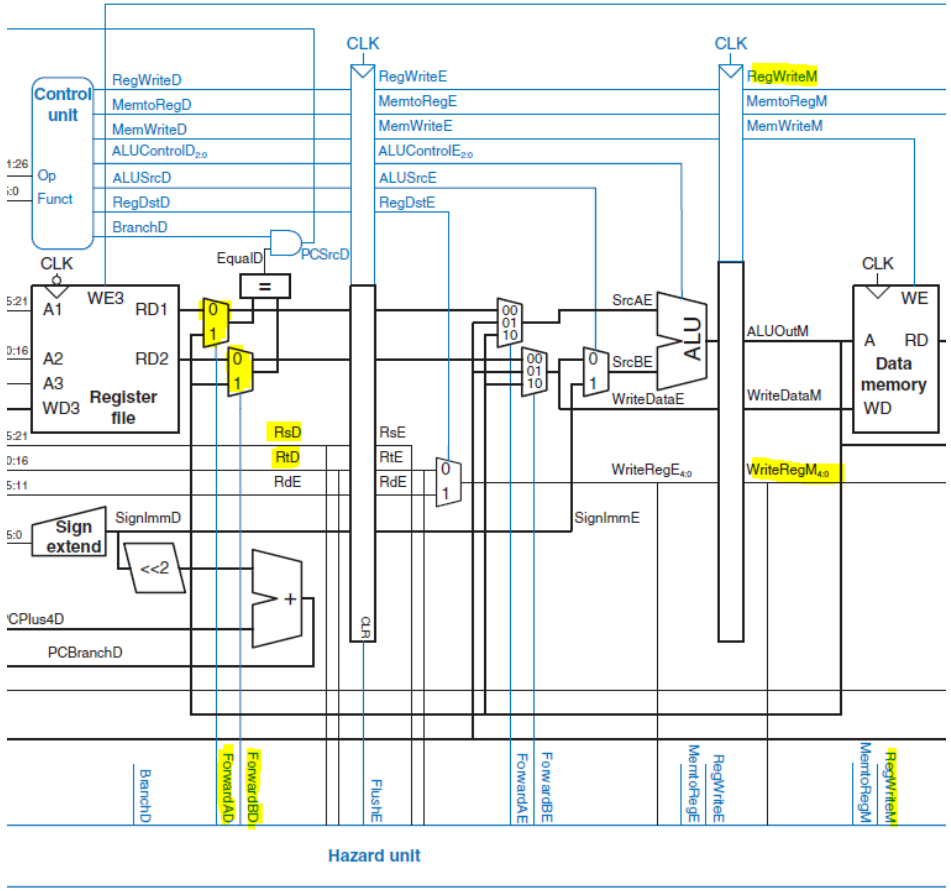


# 2. Component Implementation

- hazard_unit Implementation:
  2. Data Hazard solved by forward in Decode

```
assign ForwardAD = (RsD != 0) && (RsD == WriteRegM) && RegWriteM;

assign ForwardBD = (RtD != 0) && (RtD == WriteRegM) && RegWriteM;
```



# 2. Component Implementation

- hazard_unit Implementation:
  3. Data & Control Hazard solved by Stall

```
assign LwStall = ((RsD==RtE)||(RtD==RtE)) && MemtoRegE;

assign branchstall_cond1 = BranchD && RegWriteE && ((WriteRegE == RsD)||(WriteRegE == RtD));
assign branchstall_cond2 = BranchD && MemtoRegM&& ((WriteRegM == RsD)||(WriteRegM == RtD));

assign branchstall = branchstall_cond1 || branchstall_cond2;

assign StallIF = LwStall || branchstall;
assign StallD  = LwStall || branchstall;
assign FlushE  = LwStall || branchstall;
```

# 4. Testbench

life.augmented

# 4. Testbench

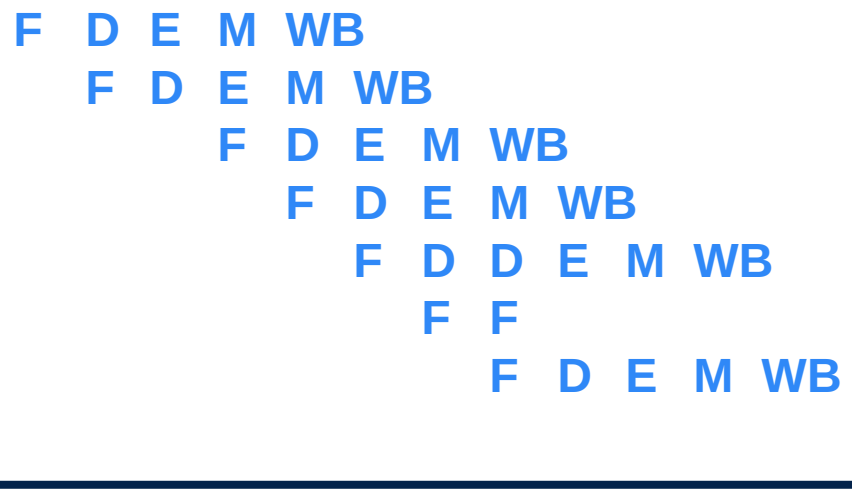
- Direct Testcase (1):

| | |
|---|---|
| addi $s0, $0, 0x0AF0 //2800 | 20100af0 |
| addi $s1, $0, 0x04B0 //1200 | 201104b0 |
| add $s2, $s0, $s1 | 02119020 |
| addi $t0, $0, 0x0FA0 //Result | 20080fa0 |
| beq $s2, $t0, NEXT1 | 12480001 |
| j ERROR | 08000016 |
| NEXT1: sub $s3, $s0, $s1 | 02119822 |
| addi $t1, $0, 0x0640 //Result | 20090640 |
| beq $s3, $t1, NEXT2 | 12690000 |
| j ERROR | 08000016 |
| NEXT2: and $s4, $s0, $s1 | 0211a024 |
| addi $t2, $0, 0x00B0 //Result | 200a00b0 |
| beq $s4, $t2, NEXT3 | 128a0000 |
| j ERROR | 08000016 |

| | |
|---|---|
| NEXT3: or $s5, $s0, $s1 | 0211a825 |
| addi $t3, $0, 0x0EF0 //Result | 200b0ef0 |
| beq $s5, $t3, NEXT4 | 12ab0000 |
| j ERROR | 08000016 |
| NEXT4: addi $t4, $0, 0x0064 | 200c0064 |
| sw $s0, 50($t4) | ae0c0032 |
| lw $t5, 50($t4) | 8dac0032 |
| beq $s0, $t5, DONE | 120d0000 |
| ERROR: addi $s7, $0, 0XDEAD | 2017dead |
| j SKIP | 08000019 |
| DONE: addi $s7, $0, 0XD08E | 2010d08e |
| SKIP: nop | 00000000 |

# 4. Testbench

- Direct Testcase (1):

| | | | | |
|---|---|---|---|---|
| **20100af0**<br>**201104b0**<br>**02119020**<br>**20080fa0**<br>**12480001**<br>**08000016**<br>**02119822** | **addi** $s0, $0, 0x0AF0<br>**addi** $s1, $0, 0x04B0<br>**add** $s2, $s0, $s1<br>**addi** $t0, $0, 0x0FA0<br>**beq** $s2, $t0, NEXT1<br>**j** ERROR<br>**NEXT1:** sub $s3, $s0, $s1 |  | Data Haz<br>Data Haz<br>Control Haz | Forward<br>Stall+Forward<br>Flush |

# 4. Testbench

- Direct Testcase (1):

| | | | | |
|---|---|---|---|---|
| **ad900032**<br>**8d8d0032**<br>**120d0002** | **sw $s0, 50($t4)**<br>**lw $t5, 50($t4)**<br>**beq $s0, $t5, DONE** | **F D E M WB**<br>**F D E M WB**<br>**F D D D E M WB** | Two Data Haz | Two Stall |

# 5. Synthesis

life.augmented

# 5. Synthesis

- Edit XDC file:
- Create clock at clk port with 100 MHz.

```
## This file is a general .xdc for the ZYBO Rev B board
## To use it in a project:
## - uncomment the lines corresponding to used pins
## - rename the used signals according to the project


##Clock signal
set_property -dict { PACKAGE_PIN L16   IOSTANDARD LVCMOS33 } [get_ports { clk }]
create_clock -add -name sys_clk_pin -period 10.00 -waveform {0 5} [get_ports { clk }]
```

# Thank you

© STMicroelectronics - All rights reserved.
The STMicroelectronics corporate logo is a registered trademark of the STMicroelectronics group of companies. All other names are the property of their respective owners.

life.augmented